

MUHAMMAD CIKAL MERDEKA

Bandung, Indonesia | +62-822-9317-4044 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [Email](#)

PROFESSIONAL EXPERIENCE

80&Company | IT Services & Consulting

AI Engineer (LLM / RAG)

June 2025 – Present

- Led comprehensive OCR accuracy optimization for automotive technical documentation, conducting systematic comparative evaluation of Azure Document Intelligence, GPT-4.1, and Gemini 2.5 Pro across complex mixed-content documents including 1200+ rows tabular data, color-coded visual elements, vehicle parts images, and multi-section layouts. Developed advanced prompt engineering strategies for vision models and designed standardized JSON and Markdown output structures optimized for downstream RAG system integration, enabling improved chunking performance for embedding models while achieving 100% data completeness and excellent accuracy in brand recognition, numerical precision, and bilingual Japanese-English technical terminology extraction.
- Architected full-stack ChatGPT-like RAG chatbot system featuring dual RAG architecture with Next.js 14, TypeScript, Prisma ORM, PostgreSQL, and Microsoft Entra ID authentication. Designed automated feedback collection pipeline and LLM-powered triage system using Azure OpenAI to replace manual spreadsheet-based workflows, contributing to system architecture design for real-time categorization and prioritization workflows, reducing feedback processing bottlenecks and improving operational efficiency.

Evermos | Connected Commerce

Data Analyst Internship

May 2024 – October 2024

- Collaborated with cross-functional stakeholders (senior data analysts and finance-accounting managers, and external auditors) to support data preparation and validation for annual financial audits.
- Automated reconciliation processes for millions of monthly AR/AP transactions data across 3+ business operations, utilizing Python, SQL, Snowflake, Google Sheets, and Metabase on a daily basis to resolve 95% of cross-system discrepancies. Additionally, gained analytics engineering experience with dbt and Apache Airflow.

Kimia Farma | Pharmaceutical & Healthcare

Big Data Analyst (Project-Based Internship)

May 2024 – June 2024

- Designed data marts and executed ETL process in BigQuery for the provided datasets.
- Developed a Looker Studio sales dashboard, identifying opportunities to boost sales by 22% in high-rated, low-transaction pharmacy branches and increase transaction volume by 15% in underperforming regions.
- Presented data insights to stakeholders, proposing strategies to increase sales of underperforming product categories by 12% through targeted marketing.

Bank Muamalat Indonesia | Banking

Business Intelligence Analyst (Project-Based Internship)

April 2024 – May 2024

- Implemented basic to advanced Excel formulas for effective data analysis in credit risks case studies.
- Executed SQL queries in BigQuery to consolidate \$1.8M in sales data across multiple product categories over a 2-year period, enhancing the bank's understanding of client business performance.
- Developed a customer analysis dashboard in Looker Studio with time series sales and quantity trends, providing insights to help our corporate client reduce churn and increase retention in top-performing categories.

FEATURED DATA & AI PROJECTS

Business Intelligence Chatbot with Langchain | [Link](#)

May 2025

- Developed an intelligent chatbot that converts natural language to SQL, enabling non-technical users to query databases instantly without manual SQL writing.
- Implemented RAG architecture using FAISS and LangChain for multi-table analysis with conversational memory and contextual follow-up capabilities.

DocuChat AI: Intelligent Document Assistant | [Link](#)

May 2025

- Built a document analysis system that extracts and retrieves information from PDF files using semantic search, enabling users to query document content through natural language conversations.
- Implemented multiple LLM integrations (GPT-4o, Claude 3.7 Sonnet, DeepSeek R1) with ChromaDB vector storage and document chunking for optimized information retrieval and scalable document processing.

Loan Prediction Based on Customer Behavior | [Link](#)

January 2025

- Enhanced credit risk assessment by analyzing business problems, gathering insights, and implementing machine learning models, while also providing valuable business recommendations to stakeholders.
- Reduced the default rate of the company from 19.55% to 7.06% and shortened the time used in manual loan processing from 6 hours to 2 minutes with a tuned random forest classification model.

Complete project list can be found on my [Portfolio Website](#).

EDUCATION & TRAININGS

Rakamin Academy

Data Science Specialization Bootcamp

October 2023 – March 2024

- Mastered core data science concepts through intensive training and applied them in comprehensive projects, covering the entire workflow from business problem definition to KPI monitoring and reporting.
- Led a team of six in developing a loan prediction machine learning application for the final project.

Bandung Institute of Technology

Bachelor of Science in Physics

August 2018 – January 2023

- Relevant coursework : Data Processing and Analysis, Statistical Physics, Simulation and Modeling of Physical Systems, Computational Physics, Experimental Physics I-III, Industrial Management, etc.
- Organizations : Physics Student Association (HIMAFI) ITB, Society of Renewable Energy, ITB Student Press

SKILLS & TOOLS

- Programming Languages : Python, SQL, R, Matlab
- Data Manipulation and Analysis : NumPy, Pandas, Polars, SciPy, Statsmodels, Regex, Excel, Google Sheets, etc
- Databases Management: PostgreSQL, MySQL, MSSQL Server, BigQuery, Snowflake, Chroma, Pinecone
- Machine Learning & AI: Scikit-learn, TensorFlow, Keras, Spacy, NLTK, Hugging Face, LangChain, Langgraph
- Data Collection & Web Scraping : BeautifulSoup, Selenium, Requests
- Data Visualization : Matplotlib, Seaborn, Tableau, Power BI, Metabase, Looker Studio, Preset
- Data Engineering & Orchestration : dbt, Apache Airflow, Dagster, Docker
- Development Tools & Environments: VS Code, Cursor, Google Colab, Git, GitHub, Streamlit